

API Security Risk Analysis Report

FUTURE_CS_03

Student Information

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Task: API Security Risk Analysis

Tools Used: Postman, GitHub

1. Introduction

Application Programming Interfaces (APIs) allow software applications to communicate and exchange data. Since APIs often expose sensitive information, they must be designed with proper security controls. This project analyzes the security of a public REST API using Postman to identify potential security risks and recommend best practices.

2. API Description

API Used: JSONPlaceholder

Base URL:

<https://jsonplaceholder.typicode.com>

JSONPlaceholder is a free REST API used for testing and learning API development. It supports common HTTP methods such as GET, POST, PUT, and DELETE.

3. Methodology

The API was tested using Postman with the following HTTP methods:

GET

POST

PUT

DELETE

The following security aspects were analyzed:

Authentication

Authorization

Input Validation

Data Exposure

Rate Limiting

4. API Testing Results

5. Security Findings

6. Recommendations

Implement secure authentication mechanisms.

Enforce proper authorization for protected resources.

Validate all client inputs before processing.

Apply API rate limiting to prevent excessive requests.

Use HTTPS for secure communication.

Monitor API logs and security events regularly.

7. Conclusion

The API testing successfully demonstrated the use of GET, POST, PUT, and DELETE requests using Postman. Although JSONPlaceholder is a testing API and intentionally lacks production security features, this analysis highlights common API security risks and recommends best practices to improve the security of real-world REST APIs.

8. Screenshots

Figure 1: GET Request (get_users.png)

Figure 2: POST Request (post_user.png)

Figure 3: PUT Request (put_user.png)

Figure 4: DELETE Request (delete_user.png)